

PRODUCT DATA SHEET

VIR VAULT (CGS / CGF)



DESCRIPTION

VIR VAULT is a Compact Laminate designed especially for Interiors (Cubicles, partitions, lockers) available for both General purpose as well as Flame retardants. Vaults are constructed by infusing multiple layers of Phenolic Kraft papers and Décor paper with Melamine resin and protected by melamine impregnated Overlay film. This compact laminate is manufactured in compliance with EN-438-4, EN-13501-1.

Description: VIR VAULT

Laminate Grade 1: CGS (Compact General Purpose Standard)

Laminate Grade 2: CGF (Compact General Purpose Fire retardant)

Sizes: 1830x1830 mm, 1830x2165 mm, 1830x3660 mm, 1830x4330 mm

Thickness: 10 mm – 18 mm

APPLICATION / APPLICATION AREA

As Cubicles, Partitions, Lockers for Office, Hotel, Airport, Public Space and Residence



FIRE PERFORMANCE (ON DEMAND)

VIR VAULT (CGF grade) is CE certified for interior applications, demonstrating compliance with the Construction Products Regulation (CPR). The product meets fire performance classification requirements as per EN 13501-1. Fire testing has been carried out in accordance with EN 13823:2020 + A1:2022 to validate its reaction to fire performance.

Reaction to Fire Classification: Classified as Class B, demonstrating excellent fire resistance with a Fire Growth Rate FIGRA0.2MJ (W/s) and FIGRA0.4MJ (W/s) of only 26.1 W/s and 18.3 W/s and a Total Heat Release (THR_{600s}) of 2.4 MJ.

Smoke Development: Rated s1, indicating minimal smoke emission. The Smoke Growth Rate (SMOGRA) is 1.5, and the Total Smoke Production (TSP_{600s}) is just 27.7 m².

Flaming Droplets / Particles: Rated d0, confirming no flaming droplets or particles ≥10 seconds within the first 600 seconds of testing.

Parameter	Requirements of B-s1,d0 Class of EN 13501- 1:2018	Observed Results	Conformity (Confirms / Do not Confirms)
A. Single Burning Item Test			
FIGRA0.2MJ (W/s)	≤120 W/s	26.1	Confirms
FIGRA0.4MJ (W/s)	≤120 W/s	18.3	Confirms
THR _{600s} (MJ)	≤7.5 MJ	2.4	Confirms
SMOGRA (m ² /s ²)	≤30 m ² /s ²	1.5	Confirms
TSP _{600s}	≤50 m ²	27.7	Confirms
LFS to edge	No LFS to edge	No	Confirms
Flaming Droplet / Particles ≥10s within the first 600 seconds	No	No	Confirms
B. Ignitability Test			
Flame to reach 150mm Mark	No Flame should reach 150mm mark	No Flame reached 150mm mark	Confirms

REACTION TO FIRE CLASSIFICATION: B-s1, d0



EN 13501
EN 13823
ISO 11925



I250326001-1



EN 13823:2010
EN ISO 11925-2:2010

INDOOR AIR EMISSIONS & VOC PERFORMANCE

VIR VAULT (CGS/CGF) is tested in accordance with EN 16516 and CDPH Standard Method v1.2 to assess formaldehyde and VOC emissions, ensuring suitability for safer indoor air quality and breathing conditions under controlled environmental conditions

Parameter	CAS No.	Test Standard	Unit	Observed Value	Limit (CDPH / EN)
Total VOC (TVOC, C5–C17)	–	CDPH v1.2 (14 days)	µg/m ³	< 3	≤ 500
Formaldehyde (EN 16516)	50-00-0	EN 16516 + ISO 16000-3	mg/m ³	0.010	≤ 0.124 (E1)
Formaldehyde (CDPH)	50-00-0	CDPH v1.2	µg/m ³	2 – 6.4	≤ 9
Acetaldehyde	75-07-0 ²	CDPH v1.2	µg/m ³	< 2 – < 5	≤ 70
Benzene	71-43-2	CDPH v1.2	µg/m ³	< 2 – < 5	≤ 1.5
Toluene	108-88-3	CDPH v1.2	µg/m ³	< 2 – < 5	≤ 150
Ethylbenzene	100-41-4	CDPH v1.2	µg/m ³	< 2 – < 5	≤ 1000
Xylenes (m-, o-, p-)	108-38-3	CDPH v1.2	µg/m ³	< 2 – < 5	≤ 350
Styrene	100-42-5	CDPH v1.2	µg/m ³	< 2 – < 5	≤ 450
Phenol	108-95-2	CDPH v1.2	µg/m ³	< 2 – < 5	≤ 100
Naphthalene	91-20-3	CDPH v1.2	µg/m ³	< 2 – < 5	≤ 4.5
Chloroform	67-66-3	CDPH v1.2	µg/m ³	< 2 – < 5	≤ 150
Carbon Tetrachloride	56-23-5	CDPH v1.2	µg/m ³	< 2 – < 5	≤ 20
Trichloroethylene	79-01-6	CDPH v1.2	µg/m ³	< 2 – < 5	≤ 300
Tetrachloroethylene	127-18-4	CDPH v1.2	µg/m ³	< 2 – < 5	≤ 17.5
Vinyl Acetate	108-05-4	CDPH v1.2	µg/m ³	< 2 – < 5	≤ 100

VIR VAULT complies with EN 16516 (E1) and CDPH Standard Method v1.2, demonstrating very low formaldehyde and VOC emissions. The product qualifies as a low-emitting material, supporting indoor air quality requirements for green building certification systems such as LEED, WELL, and BREEAM

ANTIMICROBIAL PERFORMANCE

Property	Test Standard	Test Organism / Method	Performance
Antibacterial Activity	JIS Z 2801:2010 / Amendment 1:2012	Staphylococcus aureus, E. coli, Klebsiella pneumoniae, Pseudomonas aeruginosa, MRSA, Salmonella, Enterococcus, Streptococcus	>99.99% reduction (Pass)
Antiviral Activity	ISO 21702:2019	MS2 Bacteriophage (surrogate virus)	99.26% reduction after 2 hrs / >99.99% reduction after 24 hrs
Antifungal Resistance	ASTM G21-21	Mixed fungal spore suspension	Resistant to fungal growth (Rating 0)

VIR VAULT supports hygienic surface performance through tested antibacterial, antiviral, and antifungal properties, making it suitable for applications requiring enhanced surface cleanliness in interior environments.

QUALITY CHARACTERISTICS / TECHNICAL DATA

VAULT of both General purpose and Flame- retardant (CGS/CGF) are classified under EN-438-4 and tested in accordance to EN-438-2 to verify their quality, durability, and safety.

PROPERTIES	TEST METHOD	ATTRIBUTES OF PERFORMANCE	UNIT OF MEASUREMENT	MINIMUM REQUIREMENT	RESULTS
Surface Quality	EN 438-2.4	Stain, Dirt and Similar defects	mm ² /m ²	≤ 2	1
		Fibers, hairs and stripes	mm ² /m ²	≤ 20	10
Resistance to Surface wear	EN 438-2.10	Wear Resistance - Initial point	Revolutions until IP (min)	≥ 150	450
Resistance to immersion in boiling water	EN 438-2.12	Mass increases - T≥5 mm	% (max)	2	0.4
		Thickness increases - T≥5 mm	% (max)	2	0.2
		Appearance - Gloss finish	Rating	≥ Grade 3	5
		Appearance - Other finish	Rating	≥ Grade 4	5
Resistance to water vapour	EN 438-2.14	Appearance - Gloss finish	Rating	≥ Grade 3	5
		Appearance - Other finish	Rating	≥ Grade 4	5
Resistance to Moisture	EN 438-2.15	Mass increases - T≥5 mm	% (max)	2	0.2
		Appearance	Rating	≥ Grade 4	5
Resistance to dry heat (160°)	EN 438-2.16	Appearance - Gloss finish	Rating	≥ Grade 3	5
		Appearance - Other finish	Rating	≥ Grade 4	5
Dimensional stability at elevated temperatures	EN 438-2.17	Cumulative dimensional change - T ≥ 5 mm	L % (max)	0.3	0.1
			T % (max)	0.6	0.08
Resistance to impact with large diameter ball	EN 438-2.21	T ≥ 6 mm Drop height >1800mm	Mm	Indent. Dia. 10mm	5
Resistance to Crazing	EN 438-2.24	Appearance	Rating	≥ Grade 4	5
Resistance to Scratching	EN 438-2.25	Appearance - Smooth Finishes	Rating	≥ Grade 2	5
		Appearance - Textured Finishes	Rating	≥ Grade 3	5
Resistance to Staining	EN 438-2.26	Appearance - Group 1 &2	Rating	= Grade 5	5
		Appearance - Group 3	Rating	≥ Grade 4	5
Light Fastness (Xenon arc)	EN 438-2.27	Contrast	Grey scale rating	4-5	5
	ASTM G53-91 (314-400nm)	Contrast	Wool Scale rating	≥ 4	6
Modulus of elasticity	EN ISO 178	Stress	Mpa (min)	≥ 9000	15336
Flexural strength	EN ISO 178	Stress	Mpa (min)	≥ 80	136.1
Density	EN ISO 1183	Density	gm/cm ³	≥ 1.35	1.47
Resistance to Fixing	ISO 13894-1	Screw Pullout force	N	-	3514
Thermal Conductivity	EN 12524	Thermal Conductivity	W/mK	≤ 0.30	0.2
Antibacterial Activity	JIS Z 2801: 2010	Log Reduction	%	R > 2	>99.99
Antiviral Activity	ISO 21702: 2019	Reduction rate	%	-	>99.99
Antifungal Activity	ASTM G 21 - 21	Reduction rate	Rating	≤1	0
Formaldehyde emission	EN 16516	Emission Level	mg/m ³	≤ 0.124	0.01
	EN 717-2	Classification	Class	-	E1

SURFACE FINISH

The surface finish test is conducted in accordance with EN 13722 to evaluate the angle of light reflection for each surface finish. The results are expressed in gloss units (GU), providing a precise measure of the reflective characteristics.

SURFACE	MAXIMUM DEVIATION	GLOSS UNITS
High Gloss Surface	± 15	> 70
Semi-Gloss Surface	± 10	30 - 70
Suede Surface	± 5	10 - 30
Matt surface	± 3	< 10

DIMENSIONAL TOLERANCES

NOMINAL THICKNESS (mm)	THICKNESS TOLERANCE (mm)	LENGTH & WIDTH (mm)	STRAIGHTNESS OF THE EDGES (mm/m)	SQUARENESS (mm/m)	FLATNESS (mm/m)
2-3	±0.2	+10/-0	≤ 2	≤ 1.5	≤ 2
3-5	±0.3	+10/-0	≤ 2	≤ 1.5	≤ 2
5-8	±0.4	+10/-0	≤ 2	≤ 1.5	≤ 2
8-12	±0.5	+10/-0	≤ 2	≤ 1.5	≤ 2
12-16	±0.6	+10/-0	≤ 2	≤ 1.5	≤ 2
16-20	±0.7	+10/-0	≤ 2	≤ 1.5	≤ 2
20-25	±0.8	+10/-0	≤ 2	≤ 1.5	≤ 2

